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Palaeomagnetism of the Balantak ophiolite, Sulawesi

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Abstract

Palaeomagnetic data from the East Arm of Sulawesi provide constraints on the Cretaceous and subsequent history of a fragment of the tectonic mosaic of a complex region close to the junction between SE Asia, the Indo-Australian plate and the western Pacific microplates. Primary thermoremanence has been demonstrated in samples from 23 sites in a tilted Cretaceous–Palaeogene lava sequence in the Balantak ophiolite at Batusimpang on the eastern tip of the East Arm of Sulawesi. Its direction ($D = 60.5°$, $I = -32.1°$, $\alpha_{95} = 5.5°$) indicates formation at $17 \pm 4°$S and $\sim 60°$ of subsequent clockwise rotation. The primary mode of remanence is deduced from NRM/TRM comparisons; low-temperature cycling and other rock magnetic tests point to SD magnetite as the principal remanence carrier. Supporting evidence for the palaeolatitude and northward movement of the East Arm is derived from other lavas at Binsil, and from the Boba Cherts, but lack of tectonic control limits the interpretation. The contrast between these results and the subequatorial origin of contemporary rocks on nearly Halmahera is consistent with subduction of Indian Ocean lithosphere beneath the Sunda margin in the late Mesozoic and early Tertiary. Large differences in declination of remanence in the E Sulawesi rocks indicate large clockwise and anticlockwise rotations of tectonic blocks only tens of kilometres across.

1. Introduction

The tectonically active regions between Asia and Australia are of general geological interest because they are the site of perhaps the most spectacular convergence and accretion of crust that is currently occurring anywhere on Earth. An understanding of the present-day geology and tectonics together with the unravelling of their development over the last 100 m.y. constitute a model scenario that should throw light on the evolution of older plate convergence zones. The region is large and immensely complex and the evidence is both fragmentary and obscure, and hence the whole range of available geoscientific techniques needs to be used in order to reconstruct its history. Geological evidence indicates that fragments in the tectonic mosaic were derived from Eurasia, the Indian Ocean, Australia and the Pacific and have probably been moved within this convergent zone after their amalgamation. Palaeomagnetism offers the only quantitative method of determining positions of such fragments in their contemporary palaeolatitude and longitude frame, and hence has a fundamen-

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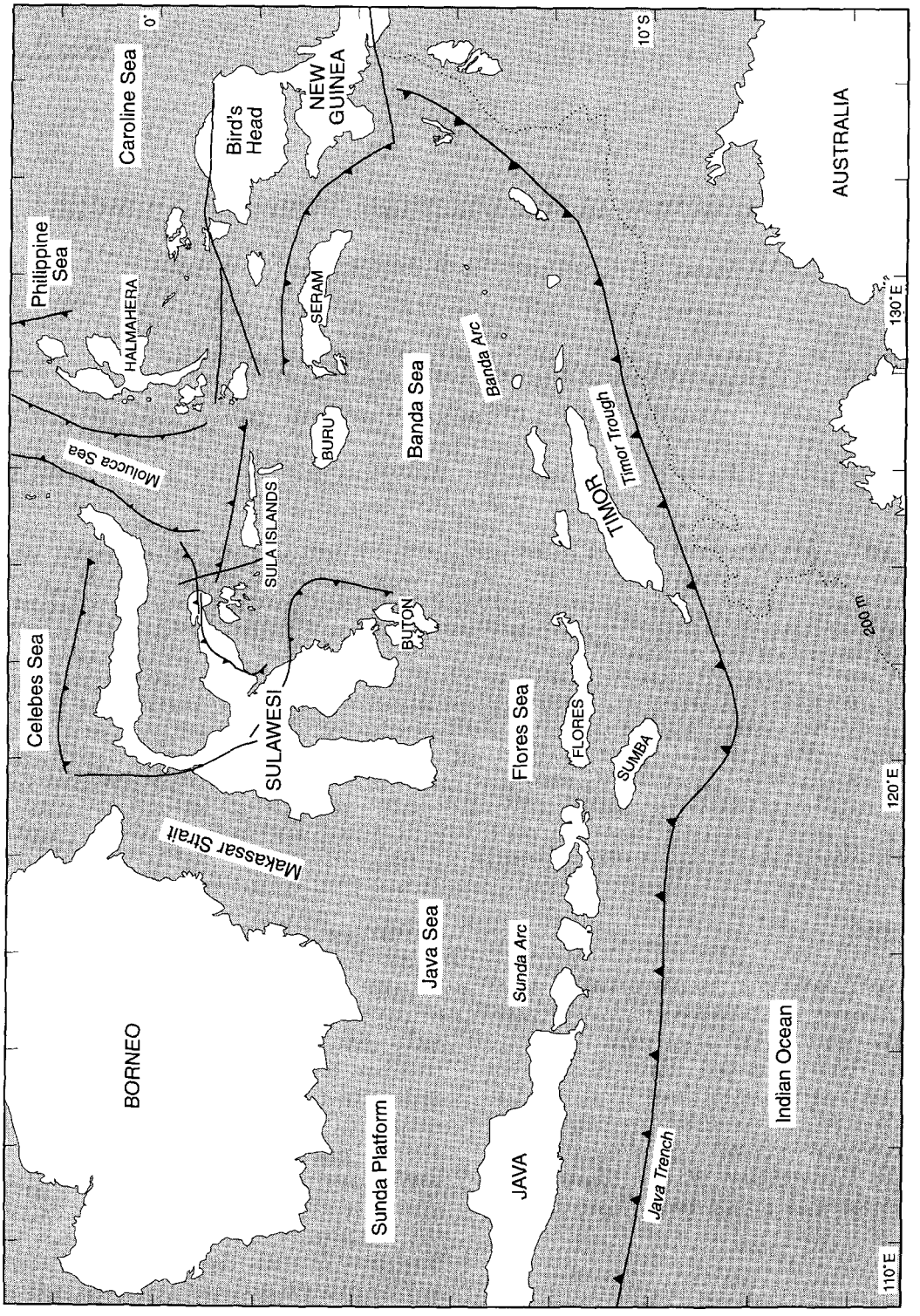


Fig. 1. The present-day tectonic situation of Sulawesi and Indonesia.

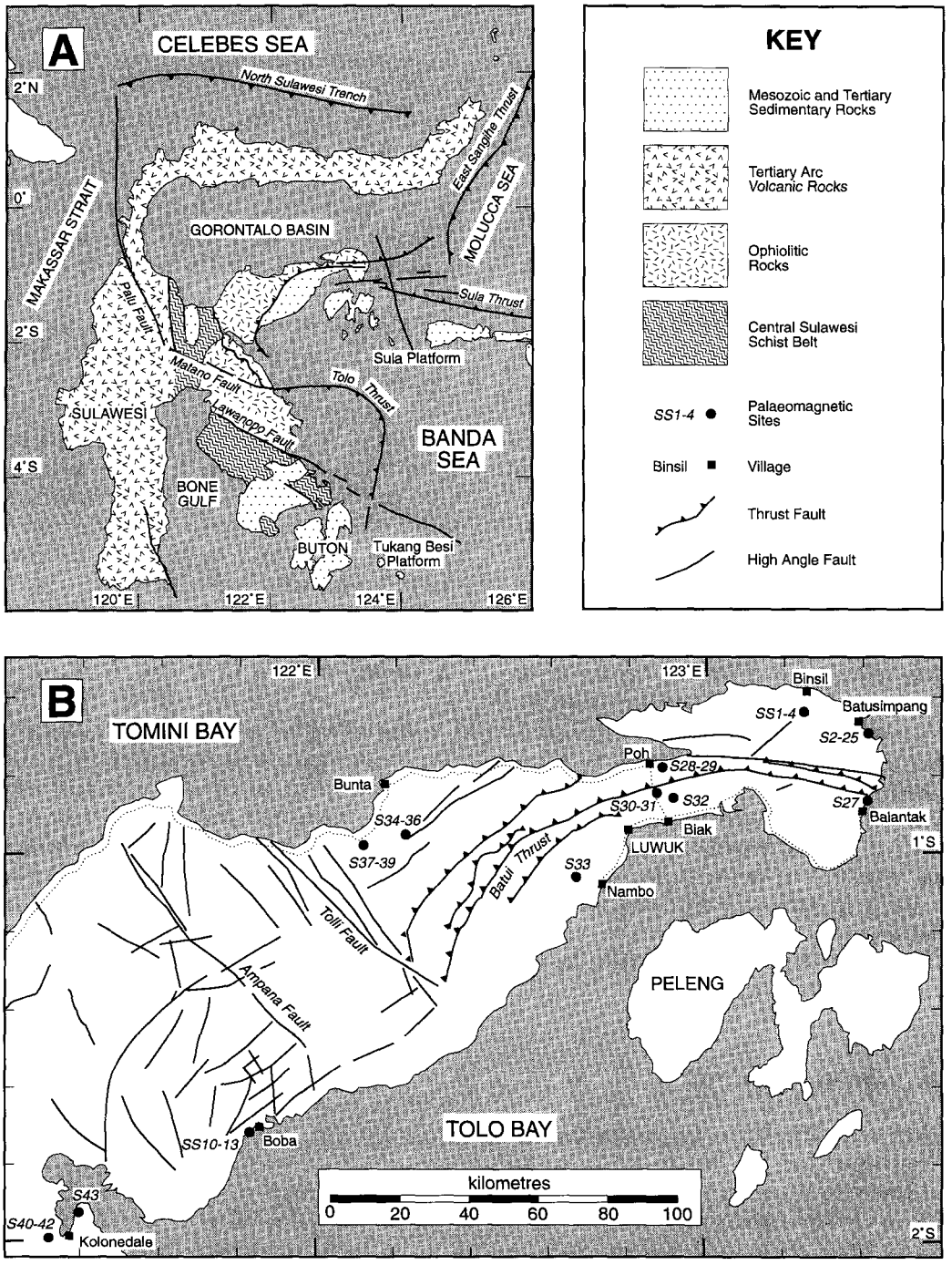


Fig. 2. (a) Tectonic elements of Sulawesi. (b) Palaeomagnetic sampling localities, East Arm of Sulawesi.

tal contribution to make. Sulawesi is located in East Indonesia close to the junction between SE Asia, the Indo-Australian plate and the western Pacific marginal microplates (Fig. 1). The region around Sulawesi includes a complex of small plates and boundaries [1] but can be considered broadly as part of Eurasia, which has a small absolute plate motion. Relative to Eurasia, Indo-Australia is currently moving NNE and the Philippine Sea Plate is moving WNW. To the south the islands of the Banda Arc separate Sulawesi from the subducting Indian Ocean lithosphere. Within the Eurasia–Pacific–Indo-Australia convergent zone Sulawesi is over-riding descending lithosphere of the Molucca Sea and the Celebes Sea and is cut by strike-slip faults linking these subduction zones to other plate boundaries. For the past 100 m.y. this part of SE Asia should be similarly described in terms of numerous small plates and plate boundaries that have shifted and changed character. Sulawesi consists of four principal tectonic belts (Fig. 2A): the West Sulawesi volcano-plutonic arc, the Central Sulawesi metamorphic belt, the East Sulawesi ophiolite belt, and the continental fragments of Banggai–Sula, Tukang Besi and Buton. This configuration has been widely interpreted in terms of collision between the eastern microcontinental fragments and the western volcanic arc [e.g., 1–4], resulting in ophiolite emplacement and metamorphism. However, as work in the region has progressed it has become clear that this apparent simplicity is partly a reflection of the incomplete knowledge of a large area which is remote, mountainous and largely covered by thick rainforest. In particular, little is known of the age and origin of the East Sulawesi ophiolite, which is likely to have formed far from its present position.

2. Previous palaeomagnetic work

There has been little previous palaeomagnetic work on Sulawesi. The earliest results were reported by Haile [5] from oriented block samples of cherts from the Southwest and Southeast Arms. They indicated that these arms were situated in different palaeogeographic regions during the Late Jurassic–Early Cretaceous, as proposed by Audley-Charles et al. [2] and Katili [3]. Data from SW Sulawesi indicate that the Southwest Arm has remained close to its present latitude since the Late Jurassic [5] and late Paleogene [6] but has rotated anticlockwise by about 45°. Directions recorded by Late Miocene volcanic rocks of the Southwest Arm are indistinguishable from those of Earth's present magnetic field. The palaeomagnetic results from SW Sulawesi are very similar to those from Cretaceous rocks of the Malay Peninsula [7] and Borneo [8], suggesting little latitudinal movement between these areas and similar anticlockwise rotations since the Late Mesozoic and before the Late Miocene. Otofuji et al. [9] suggested that the North Arm of Sulawesi has rotated clockwise by more than 90° with no significant latitude change since the Eocene–Early Miocene, but their data are insufficient to justify any such conclusion. ODP Leg 124 drilling in the Celebes Sea indicated no significant difference from present-day latitude, but about a 60° anticlockwise rotation between the late Eocene and Early Miocene.

3. Geology

Ophiolites are sequences of basic and ultrabasic rocks which, when complete and intact, consist of a lower ultramafic complex overlain by layered cumulates and gabbros, sheeted dykes and a sequence of lavas. They are widely interpreted as fragments of former oceanic lithosphere, although the tectonic setting of these relicts is controversial, and there is increasing evidence, principally from geochemistry, that many ophiolites formed in an arc-related environment. The East Sulawesi ophiolite belt extends discontinuously from the East Arm, through Central Sulawesi, into the Southeast Arm and Buton, and the ophiolite belt forms part of a structurally complex zone interpreted by Hamilton [1] to be the result of W-dipping subduction during the Late cretaceous–early Tertiary. The Balantak ophiolite is the name given to the ophiolitic rocks of the East Arm where there is a broadly complete, but dismembered and highly faulted, se-

quence. The geological summary given here is based on Simandjuntak [10], who divides the ophiolite into four parts thought to represent ascending parts of the ophiolite sequence: peridotites, minor gabbros and diorites, dolerites and basalts. Ultramafic rocks are predominantly noncumulate harzburgites and subsidiary dunites with local cumulates. They occur as faulted, usually imbricated, slivers which are largely serpentinised. Gabbros include cumulate, isotropic and pegmatitic varieties, locally associated with troctolites, anorthosites and trondhjemites. A dyke complex is faulted against gabbros on the east coast of the East Arm and field relationships suggest this is above gabbros and below pillow lavas. The dykes are 5–100 cm thick and trend mainly NW–SE with variable but steep dips to both the northeast and the southwest. Silver et al. [4] reported NNW–SSE trending sheeted dykes along tributaries of the Binsil River. The thickness of the dyke complex is uncertain, due to tectonism: profiles measured perpendicular to the sheeting suggest up to 3 km but this may be an over-estimate due to repetition by faulting. The lava sequence is dominated by pillow basalts with subsidiary massive lavas, thin lava flows, hyaloclastites and brecciated basalts. In some areas there is a transition to the sheeted complex. On the coast of the East Arm there is a faulted contact between sheeted dykes and basalts. The lavas are highly faulted and it is difficult to determine their thickness; if there is no repetition there may be up to 3 km of lavas in places. Most of the basalts are porphyritic with phenocrysts of plagioclase and pyroxene in a glassy matrix containing microlites of plagioclase, pyroxene and magnetite and they commonly contain amygdales up to 2.5 cm across, filled by zeolites and/or calcite. The basalts are typically highly altered with plagioclase replaced by zeolites, saussurite and carbonate; pyroxene is replaced by chlorite, epidote, iron oxides and rarely hornblende; olivine is altered to serpentine; and glass is partially devitrified. In parts of the East Arm sedimentary rocks are tectonically associated with ophiolitic rocks. There is locally calcilutite infilling of pillow interstices and in some areas lavas or hyaloclastites are overlain by lenticular calcilutites; all are unfossiliferous. In the Boba River, cherts and limestones are assigned to the Boba Beds, which comprise an informal subdivision of the Matano Formation of Sukamto and Simandjuntak [11]. They are largely thinly bedded radiolarian cherts with calcilutites; in places the cherts contain MnO in lenses and nodules up to 3 cm thick and ~ 1 m long, and in fractures. Both the cherts and the limestones contain microfossils. Their thickness is uncertain due to tectonism, but it is estimated to be up to 50 m. The Boba Beds are tightly folded, faulted and thrust at all scales, and bedding attitudes change rapidly over short distances.

4. Palaeomagnetic sampling

The Boba Beds were sampled in the Boba River (Fig. 2B) over a distance of about 3 km where cherts are exposed in small discontinuous outcrops with different bedding attitudes in individual exposures. Lavas were sampled at Poh Head at the east end of the East Arm where there is an extensive area of ophiolites dominated by basaltic lavas. Near Batusimpang village (Fig. 2B) there is a well-exposed continuous coastal section of several kilometres in length that is dominated by pillow lavas which are the right way up, young northwards and pass down into sheeted dykes. The section includes layered and massive lava flows and dips NNW at about 30°. Twenty-four sites were drilled in lava flows (only two of which were in pillowed lavas) representing an estimated stratigraphic thickness of more than 500 m, assuming no repetition by faulting. Four sites were drilled in pillow lavas near Binsil village in the Bombon River valley (Fig. 2B). The pillow lavas young northwestwards, but their dip is uncertain due to the lack of layering and interbedded sediments, which precludes adequate tectonic correlation of the remanence directions calcilutite and hyaloclastic material fills in the pillow interstices.

5. Age of the ophiolite and associated rocks

Stratigraphic relationships between the ophiolite and the Boba Beds, and between the cherts

and limestones of the Boba Beds are not clear due to tectonism. A single stratigraphic contact between lavas and undated cherts at one locality near Kolonedale suggests that the unit was deposited on top of the ophiolite [10]. Biostratigraphic dating of the Boba Beds indicates that the cherts are Valanginian–lower Cenomanian (based on radiolaria) and the calcilutites are Upper Cretaceous (based on planktonic foraminifera). There are some K-Ar age data from the ophiolite: Six K-Ar determinations on Batusimpang lavas made during this study yielded a range of ages from 79 to 16 Ma. Field mapping suggests that all the Batusimpang lavas are likely to be of similar age and XRF whole-rock analyses suggest that Neogene ages reflect low-temperature alteration. Excluding these, one sample has an age of 79 ± 5 Ma and the remaining three samples range from 60 to 54 Ma. Ten K-Ar ages from Binsil and Batusimpang reported by Simandjuntak [10] include gabbros with age ranges from 93 to 48 Ma and four basalts which range from 54 to 38 Ma. Simandjuntak [10] interprets the Boba cherts as the sedimentary cover deposited at or near an Early Cretaceous spreading centre, although the sequence of older chert and younger limestone in the inferred sediment cover is unusual for normal ocean floor where the mid-ocean ridge is initially above, and later subsides below, the carbonate compensation depth. If the interpretation of a Cretaceous age for the ophiolite is correct the oldest K-Ar ages would be the closest to the formation age of the basalts, and the younger K-Ar ages may reflect argon loss due to alteration. Supporting this suggestion, the oldest K-Ar age of 93 ± 2 Ma from an ophiolite gabbro is close to the minimum age (Cenomanian) of the Boba cherts [12]. However, as noted above mapping suggests that all the Batusimpang lavas are of similar age. Most K-Ar ages from the Batusimpang and Binsil lavas are Tertiary and other possible interpretations of these ages are

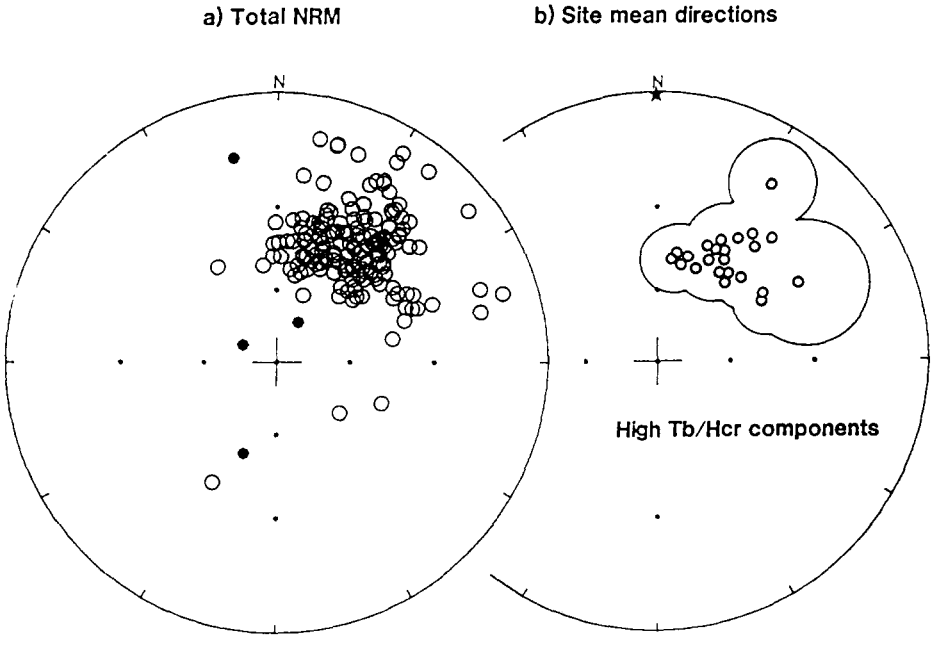


Fig. 3. Palaeomagnetic directions in the Batusimpang lava basalt flows. (a) Total NRM directions of 191 samples, in situ. (b) Site mean directions in situ of high-T_B and high-H_{CR} components with the envelope of circles of 95% confidence. Stereographic projections: open/closed symbols represent projections on the upper/lower hemisphere. The present-day geomagnetic field direction is marked by a star in (b).

that the ophiolite is composite with early Tertiary lavas imbricated with Cretaceous gabbros and cherts or that the entire ophiolite is early Tertiary in age. Simandjuntak interprets the younger K-Ar ages to represent either ocean floor segments or seamounts of Tertiary age. Thus, the East Arm

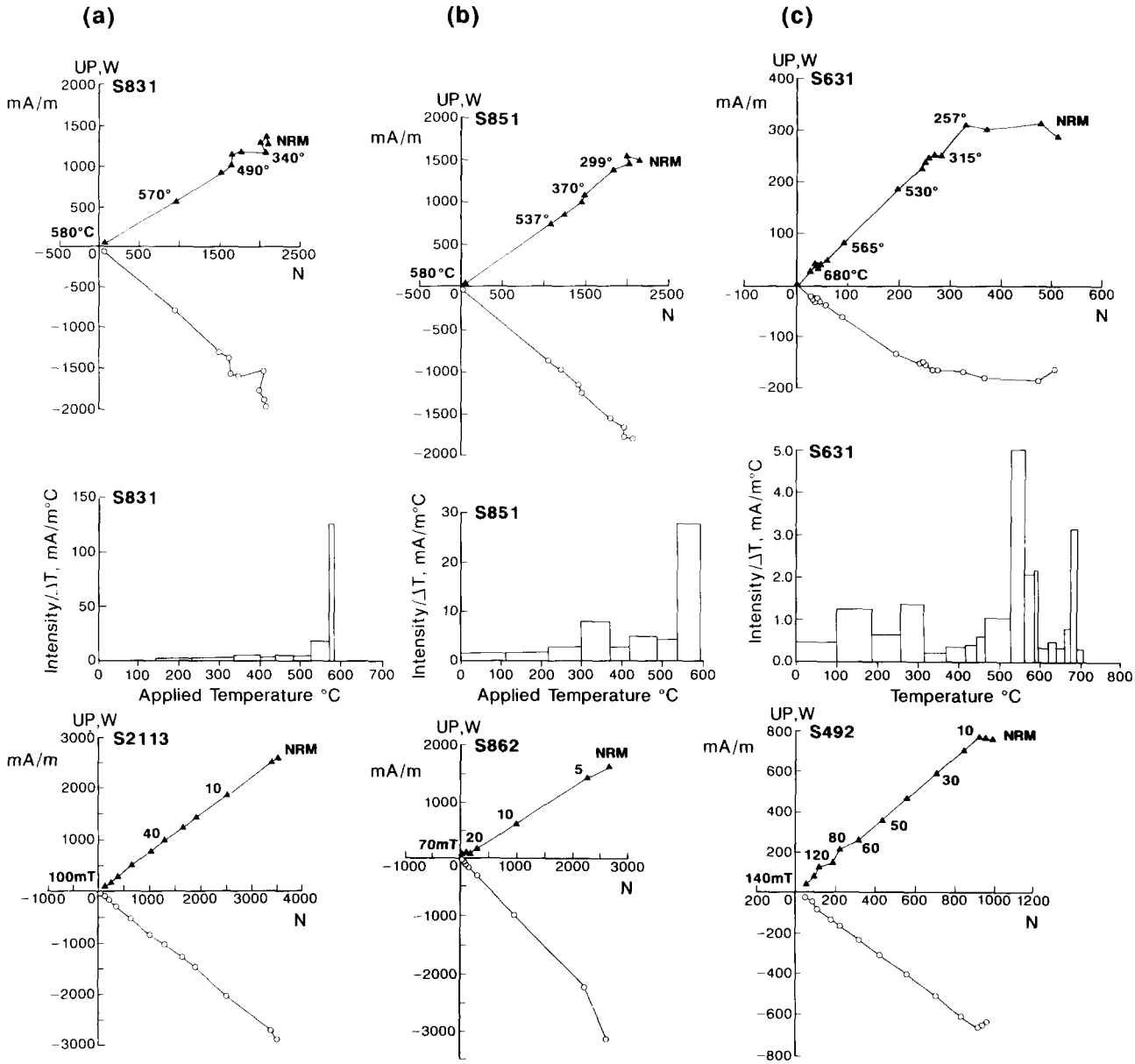


Fig. 4. Demagnetization characteristics of Batusimpang basalts. (a) Group B. (b) Group A. (c) Group C. Top and middle rows: Thermal demagnetization of samples 831, 851 and 631. Bottom row: af demagnetization of samples 2113, 862 and 492. The top and bottom rows are orthogonal plots, with circles referring to the horizontal projection and triangles to the vertical. The middle row shows T_B spectra measured along the demagnetization trajectory; the ordinate is normalised to magnetisation per degree celcius to eliminate distortion due to the different temperature increments ΔT.

lavas sampled in this study are possibly Cretaceous (Cenomanian) or Tertiary (Eocene). More age data and trace element geochemical data are needed to assess these interpretations.

6. Batusimpang lavas: Palaeomagnetic data

The NRM of 191 samples from 24 sites in the Batusimpang lavas was measured using a CCL cryogenic magnetometer. The directions are strongly concentrated upwards to the north-northeast (Fig. 3a). Intensities range between 0.05 and 9.3 A m^{-1} and about 40% are less than 1 A m^{-1} The magnetic susceptibilities range from 0.01 to 0.63. One hundred and forty of these samples have been progressively demagnetized using alternating field (af) and thermal methods at between 10 and 20 steps up to 200 mT or 600°C or more. Two af demagnetisers were used: a two-axis tumbling instrument following the design of McElhinny [13], and a single-axis demagnetizer (for which three orthogonal treatments are needed at each applied field) built by G. McCallum following a design by D.R. Watts. Two thermal demagnetizers were also used, the first having field cancellation by orthogonal coil systems with feedback circuitry and the second using multiple μ-metal shielding. In both furnaces the field was reduced to less than 5 nT. For some of the work, a nitrogen atmosphere was used to suppress oxidation in the samples during heating

Table 1
Site mean analysis of high T_B and H_{CR} components in the Batusimpang lavas (in situ)

Site	*N*	*Dec*	*Inc*	*R*	*k*	α_{95}
S2	7	22.3	−38.9	6.9669	181.4	4.5
S3	8	24.2	−39.8	7.9618	183.1	4.1
S4	6	29.3	−37.1	5.8184	27.5	13.0
S5	7	41.3	−32.0	6.9569	139.1	5.1
S6	6	38.8	−28.8	5.8912	46.0	9.9
S7	5	62.3	−28.3	4.7586	16.6	19.4
S8	7	43.0	−25.8	6.9544	131.7	5.3
S9	6	32.9	−14.5	5.8579	35.2	11.4
S10	7	37.3	−29.3	6.9149	70.5	7.2
S11	6	34.7	−41.8	5.9626	133.5	5.8
S12	5	29.1	−43.8	4.8469	26.1	15.2
S13	6	35.4	−32.8	5.8296	29.3	12.6
S14	6	61.3	−41.3	5.8512	33.6	11.7
S15	6	36.5	−45.9	5.8788	41.3	10.6
S16	5	58.8	−40.4	4.9818	219.9	5.2
S18	5	44.5	−49.4	4.9873	315.8	4.3
S19	6	40.9	−46.5	5.9159	59.4	8.7
S20	6	46.6	−42.7	5.8672	37.6	11.1
S21	6	15.2	−51.3	5.9008	50.4	9.5
S22	5	17.8	−48.0	4.9829	234.6	5.0
S23	5	11.8	−45.7	4.9239	52.6	10.7
S24	5	9.1	−48.5	4.8969	387.9	12.4
S25 *	5	294.1	20.5	4.9729	148.0	6.3
S26	4	26.6	−50.2	3.9757	123.6	8.3
B = 23	135	35.4	−40.1	22.306	131.7	5.5
After tilt correction (strike 215°, dip 34°NW)						
B = 23	135	60.5	−32.1 +	22.3061	31.7	5.5

* This datum is not used in the calculation of the overall means. + The sampling locality is 0.66°S, 123.39°E and the palaeomagnetic pole corresponding to the mean direction after tilt correction is 28.2°N, 232.9°E, $d_p = 3.5°$, $d_m = 6.2°$.

and cooling, but it emerged that this was not necessary. Magnetic susceptibility was monitored at each stage using a Molspin MINISEP susceptibility bridge.

The directions within the demagnetisation trajectory were analyzed using the LINEFIND algorithm [14]. It is useful to distinguish three types of remanent magnetization distribution in the Batusimpang lavas according to whether they have one, two or three peaks in their blocking temperature spectrum (apart from a small viscous component aligned with the present geomagnetic field). These are referred to as Groups B, A and C respectively. Group B (Fig. 4a) has a single T_B peak between 500 and 580°C and NRM intensities between 1 and 4.5 A m^{-1}. Group A (Fig. 4b) has two T_B ranges between 200–400 and 500–580°C. Group C (Fig. 4c) has a third peak between 600 and 680°C. Group A tends to have the strongest NRM intensities (2–9 A m^{-1}) and the lowest coercivities (up to 70 mT); Group C tends to have the lowest intensities (less than 1.5 A m^{-1}) and the highest coercivities (greater than 100 mT). These features are all illustrated in Fig. 4.

Generally a single large directional component of magnetization was obtained after removing the small viscous remanence directed towards the present Earth field, after treatments of 20 mT or 300°C or less. Thus all the ancient remanence appears to have been acquired in the same magnetic field and hence at the same time. The only clear exceptions to this were in a single sample from site 3 and three from site 12. These all have Group C blocking temperature distributions, the higher and lower ranges carrying the characteristic magnetization, but the intermediate range carrying an approximately reversed component. In the example from site 3 the T_B range of the intermediate component is 350–490°C; in the examples from site 12 it reaches as high as 587°C.

The directional statistics of the characteristic

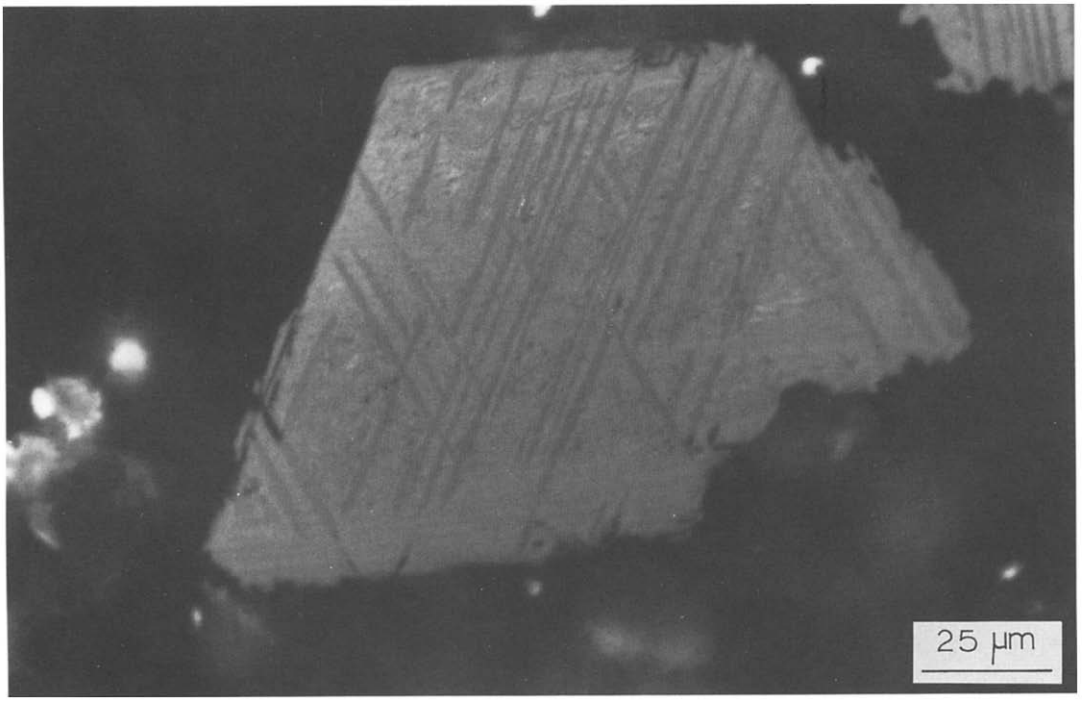


Fig. 5. Optical photomicrograph showing euhedral, titanium-poor magnetite grain from the Batusimpang lava flows. Dark grey ilmenite can be seen in the form of high-temperature exsolution lamellae, providing evidence of deuteric high-temperature thermoremanence.

magnetization are given in Table 1 and illustrated in Fig. 3b. All of these sites are in lava flows, except for sites 25 and 26 which were in pillows. Site 26 has Group A blocking temperature characteristics and similar directional properties to the main flow sequence. Site 25, in contrast, has a soft viscous remanence plus the single T_B spike 500–580°C characteristic of Group A, although its directions are anomalous (northwest and downwards). We do not have a complete explanation for this anomaly: local faulting at this site cannot alone account for it, but could have provided fluid conduits for later alteration effects. This site has been excluded from the mean statistics in Table 1.

7. Batusimpang lavas: Origin of remanence

The three blocking temperature ranges in the Batusimpang lavas suggest the presence of hematite, magnetite and either titanium-rich magnetite or maghemite. Skeletal titanomagnetites up to 180 μm in diameter are clearly identifiable in reflected light. Their form is characteristic of rapidly cooled volcanic rocks and they are inferred to be primary. Exsolution of ilmenite within titanomagnetite as in Fig. 5 is a deuteric high-temperature phenomenon (oxidation class 3 in the classification of Wilson and Watkins [15]). Exsolution leads to composite grains with compositions close to the magnetite and ilmenite end members, and this has been confirmed by electron probe microanalysis (e.g., at site 6). The range of lower blocking temperatures could either be due to the wide range of grain sizes or to the presence of unexsolved high-titanium titanomagnetites. The sample from site 12 that has three T_B peaks (Group C) contains hematite, pure magnetite and also titanomagnetite with a Ti:Fe ratio of ~ 0.2, which would be expected to have blocking temperatures around 400°C.

The analytical electron microscopy available to us does not have the resolution to analyse micrometre-size grains or smaller, which is critical because of the importance of single-domain (SD) magnetite as a likely remanence carrier. To investigate the presence of this magnetite we subjected representative specimens to low-temperature cycling in magnetic field-free space between the laboratory temperature and liquid nitrogen temperature (−196°C). Typically after two cycles the viscous remanence is removed and the remanence has shifted into the characteristic direction (alternating field demagnetization confirmed this to be univectorial (Fig. 6)).

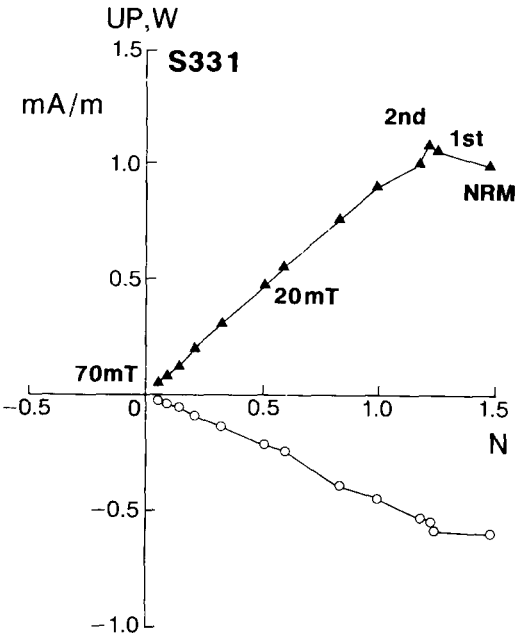


Fig. 6. Orthogonal plot showing af demagnetization characteristics of sample S331 following two cycles of cooling to liquid nitrogen temperature. Open symbols for horizontal projection, solid symbols for vertical.

Fig. 7 illustrates the recovery of NRM during warming from liquid nitrogen temperature. To measure this the sample was quickly transferred from the nitrogen dewar to a Molspin magnetometer at room temperature. In the absence of any means to control or measure the temperature during recovery, the sample was simply spun about its X-axis and the other two orthogonal components were monitored every ~ 10 s. In all but one of the tested samples the drop in intensity during low-temperature cycling was less than 30%, and no more than 1 or 2% recovery was observed during the warming phase. This indicates the dominant contribution of SD magnetite to the

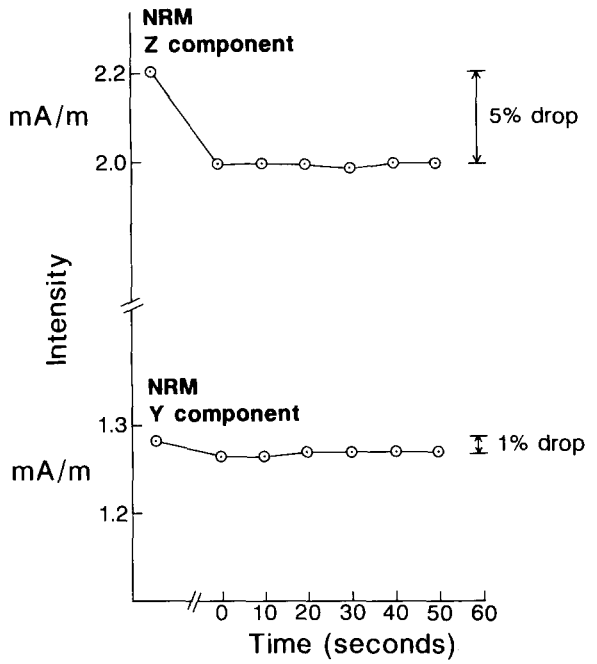


Fig. 7. NRM intensity in sample S891 from Batusimpang upon cooling to liquid nitrogen temperature. About 5% of the moment is lost upon initial cooling. There is negligible recovery of NRM during warming back to laboratory temperature over the subsequent minute.

remanence [16,17]. The exception to this behaviour was sample 12.61, a sister to one of the Group C samples with anomalous remanence (see section 3); in this case the monitored components changed sign and the eventual intensity drop was 44%.

The further observation that IRM saturates at $\leqslant 100$ mT, and with backfield coercivities of 25–45 mT, also shows that magnetite dominates the bulk magnetic mineralogy of the Batusimpang lavas.

Finally, comparisons of the thermal demagnetization properties of NRM and TRM [18] were made in order to assess the likelihood of the NRM being of thermoremanent origin. A laboratory field of 60 μT was used, with a nitrogen atmosphere to suppress alteration. All samples behaved isotropically: TRM was always along the applied field. Susceptibility was monitored throughout and no significant changes were observed until heating had proceeded beyond 600°C.

Samples with NRM blocking temperature characteristics in Groups A and B showed a good match between NRM and TRM either throughout the whole T_B range (Fig. 8a) or above ~ 300°C (Fig. 8b). In both these cases the characteristic remanence, or at least the higher T_B part of it, is inferred to be primary TRM. A sample from Group C did not conform to this behaviour (Fig. 8c) and it is likely that most if not all of its NRM is CRM. Group C occurs predominantly in the lower, eastern, part of the sequence, suggesting a region of alteration of magnetic mineralogy. Our inference of primary TRM origin allows us to choose the tilt-corrected remanence direction for our tectonic analysis ($D = 60.5°$, $I = -32.1°$, $\alpha_{95} = 5.5$).

These experiments were not designed to provide a reliable palaeointensity estimate. Only the five samples with Group B NRM showed sufficiently precise linearity of NRM/TRM plots, but it is worth mentioning that these yield a palaeointensity estimate of 14 ± 4 μT, only about 30% or the present field at this latitude.

8. Binsil lavas

The NRM of the Binsil pillows is upwards to the northwest, except for site SS4 which has southwesterly NRM (Table 2), with intensities in the range 0.25 to 5.18 A m^{-1}. The coercivities of remanence are low (< 30 mT) and blocking temperatures are almost entirely in the range 200–400°C. Within-site precisions are improved by af and thermal cleaning and directions move away from the present Earth field (Table 2). The remanence is consistent with the optical observation of oxidised titanomagnetites (Fig. 9) with low-temperature oxidation characteristics. The declination difference at site SS4 is probably due to rotation between local faults and this site is not included in the mean statistics in Table 2. Unlike the situation at Batusimpang, we have no decisive evidence for the age of the remanence and, in the absence of precise measurements of tectonic dip, we can only regard the data as tentative evidence for anticlockwise rotation and an origin in a low southerly latitude.

9. Boba cherts

The cherts are much more weakly magnetized than the igneous rocks (0.1–30.5 mA m^{-1}). Non-random remanences were found at only four of the eight sites that were sampled. The NRMs are shallow and grouped slightly west of the present Earth field direction (Fig. 10). Thermal demagnetization (Fig. 11) confirms that a large low blocking temperature component dominates the remanence and that when it is removed well-grouped northwesterly negative directions are isolated (Table 3 and Fig. 10). The T_B range of this component is 400–600°C, and drops sharply around 575°C suggesting near-pure magnetite as the main remanence carrier. A minority of samples have a maximum T_B as high as 650°C, showing that hematite is also a carrier in them. In these cases the two carriers are magnetized colinearly, suggesting that deposition, alteration and remanence acquisition were closely comtemporaneous. In the absence of sufficient variation in dip to enable us to relate the remanence to folding, there is no decisive evidence that the NRM is primary.

10. Tectonic implications

The most important result of this study is the stable remanence direction found in the Batusimpang lavas ($D = 60.5°$, $I = -32.1°$, $\alpha_{95} = 5.5°$). This direction corresponds to a palaeolatitude of 17 ± 4°S and hence about 17° of northward migration accompanied by ~ 60° of clockwise rotation since the remanence was acquired. This remanence has been shown to be syngenetic with lava formation. The palaeolatitude 16.7 ± 14.0°S calculated from the inclinations in the Binsil lavas is consistent with this, but its significance is uncertain because it is not clear if the Binsil sequence was magnetised before or after tectonic deformation and there is no estimate of dip at Binsil to make a structural correction if that were appropriate. If the remanence in the Boba cherts is of primary, early Cretaceous origin its direction when tilt corrected ($D = 285.9°$, $I = -38.9°$, $\alpha_{95} = 9.3°$) would indicates that it originated in the

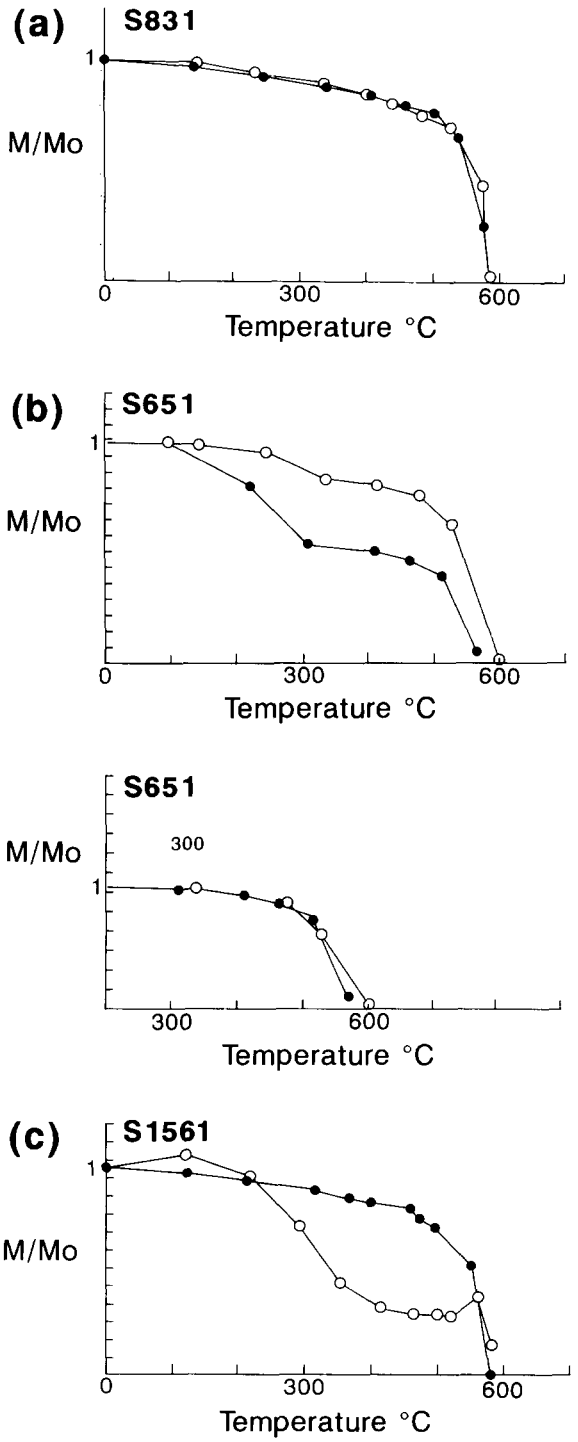


Fig. 8. Normalised thermal demagnetization curves of NRM and TRM in samples from (a) Group B, (b) Group A and (c) Group C. The lower curve in (b) is normalised to 300°C; all others are normalised to room temperature.

Table 2
Remanence directions in situ of Binsil pillow lavas (locality 0.6°S, 123.4°E)

Site	Total NRM					High-T_B, high-H_{CR} components				
	N	Intensity (mA m^{-1})	$D°$	$I°$	k	N	$D°$	$I°$	k	α_{95}
SS1	8	3054–674	336.1	−43.1	4.9	7	326.6	−35.5	34.6	10.4
SS2	6	5182–966	355.5	−15.9	17.9	6	347.3	−18.0	81.0	7.5
SS3	9	2416–859	333.4	−40.8	13.3	7	331.8	−38.3	18.2	14.6
SS4 *	8	1157–250	259.9	−28.6	2.1	9	233.5	−45.9	9.8	17.3

Mean B = 3 sites, $D = 335.9°$, $I = -30.9°$, $R = 2.9362$, $k = 31.4$, $\alpha_{55} = 22.4$, pole 61.3°N, 358.0°E, $d_p = 14.0$, $d_m = 25.0$

* This datum is not used in the calculation of the overall mean.

southern hemisphere (22 ± 7°S) and therefore has undergone about 22° of northward movement and a 74° anticlockwise rotation.

We cannot exclude the alternative scenario that the Boba chert NRM was not fixed until the rocks had been tilted into their present position, in a palaeolatitude of 9°S (corresponding to $I = -17.5°$): If this happened after the lavas were magnetised, this scenario too is consistent with progressive northward translation of the East Arm.

Irrespective of whether the East Arm lavas are Cretaceous or Tertiary in age, an important conclusion is that all the rocks of the East Sulawesi ophiolite have a clear southern hemisphere origin and have undergone large rotations, either clockwise or anticlockwise, since their formation. The palaeolatitudes are in marked contrast to those of

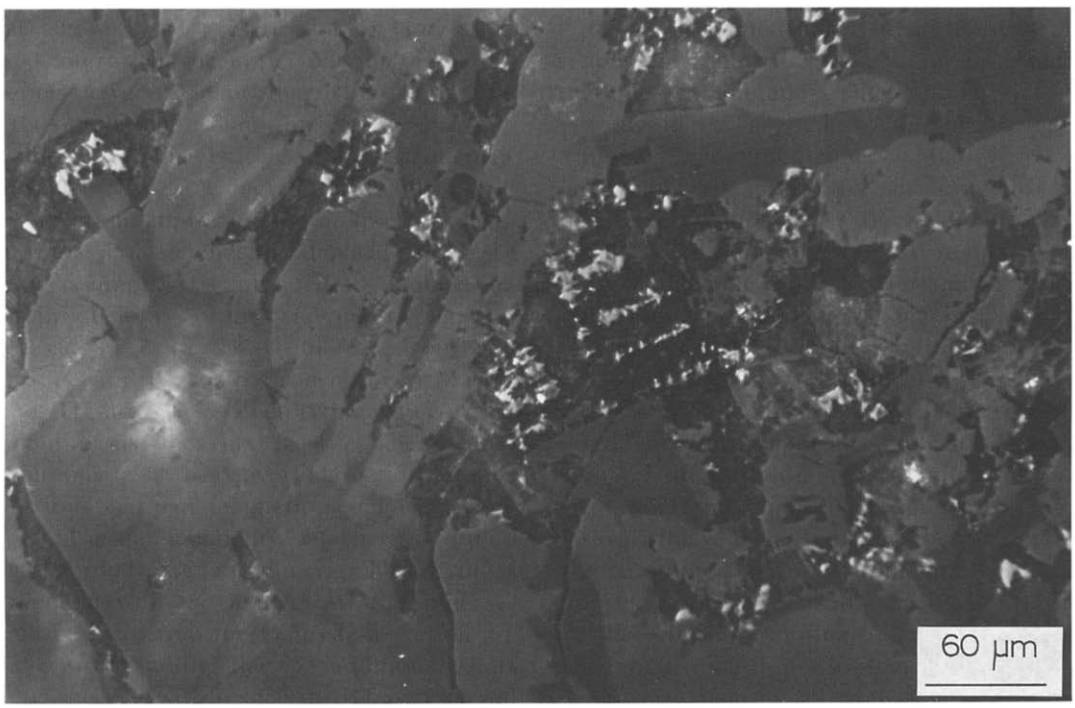


Fig. 9. Reflection photomicrograph of Binsil pillow basalt showing altered titanomagnetite grains, in which the residual titanomagnetite blebs are up to ~ 10 μm in diameter.

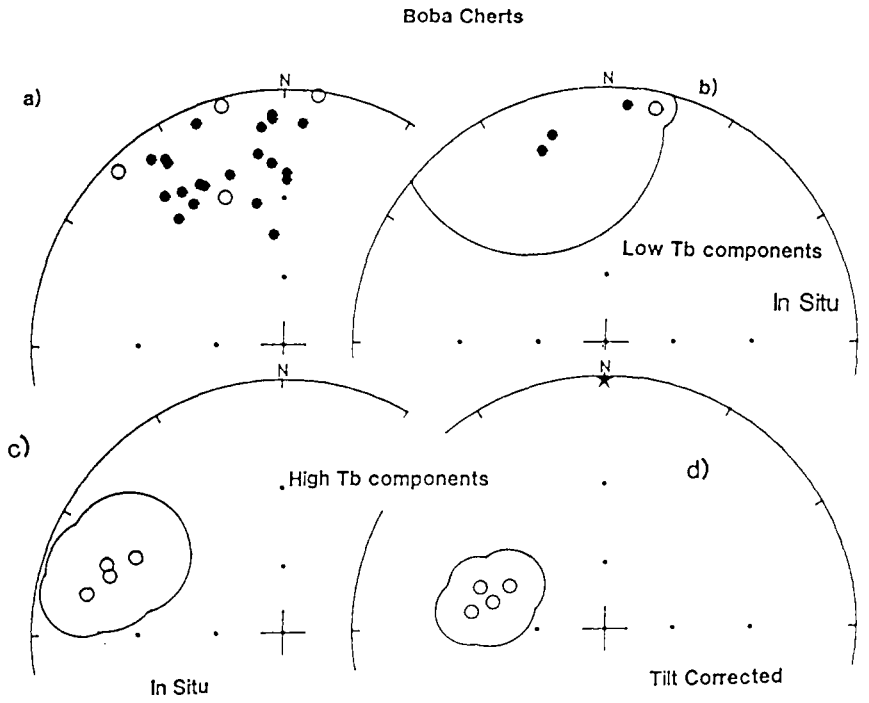


Fig. 10. Palaeomagnetic directions in Boba cherts (stereographic projection). (a) Sample NRMs in situ. (b) Site means of low-T_B components in situ. (c) and (d) Site means of high-T_B components in situ and tilt corrected respectively. Circles of 95% confidence are indicated.

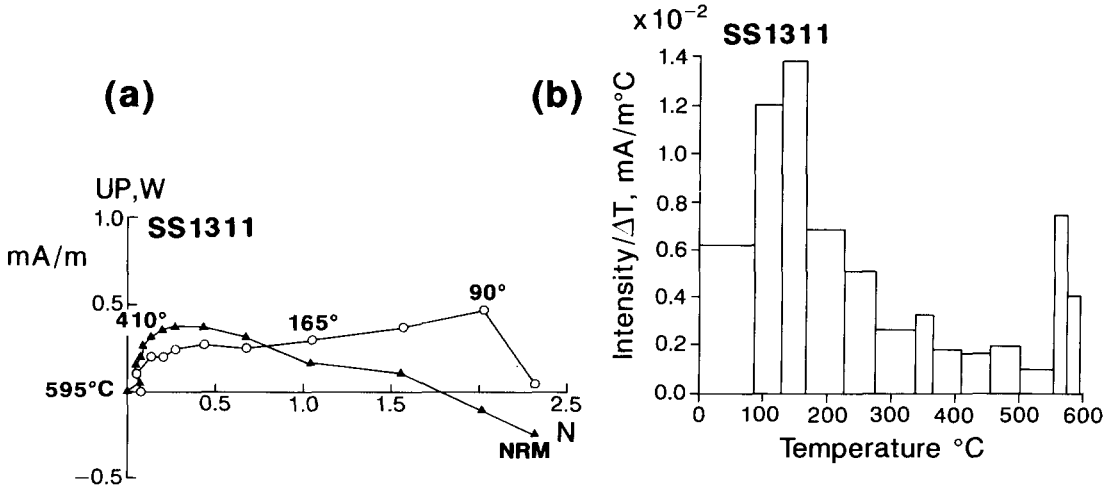


Fig. 11. Thermal demagnetization of sample SS1311 from Boba cherts. (a) Orthogonal projection showing a gentle northerly component with T_B between 90 and 410°C and a westerly component with higher T_B. Blocking temperature spectrum is shown in (b).

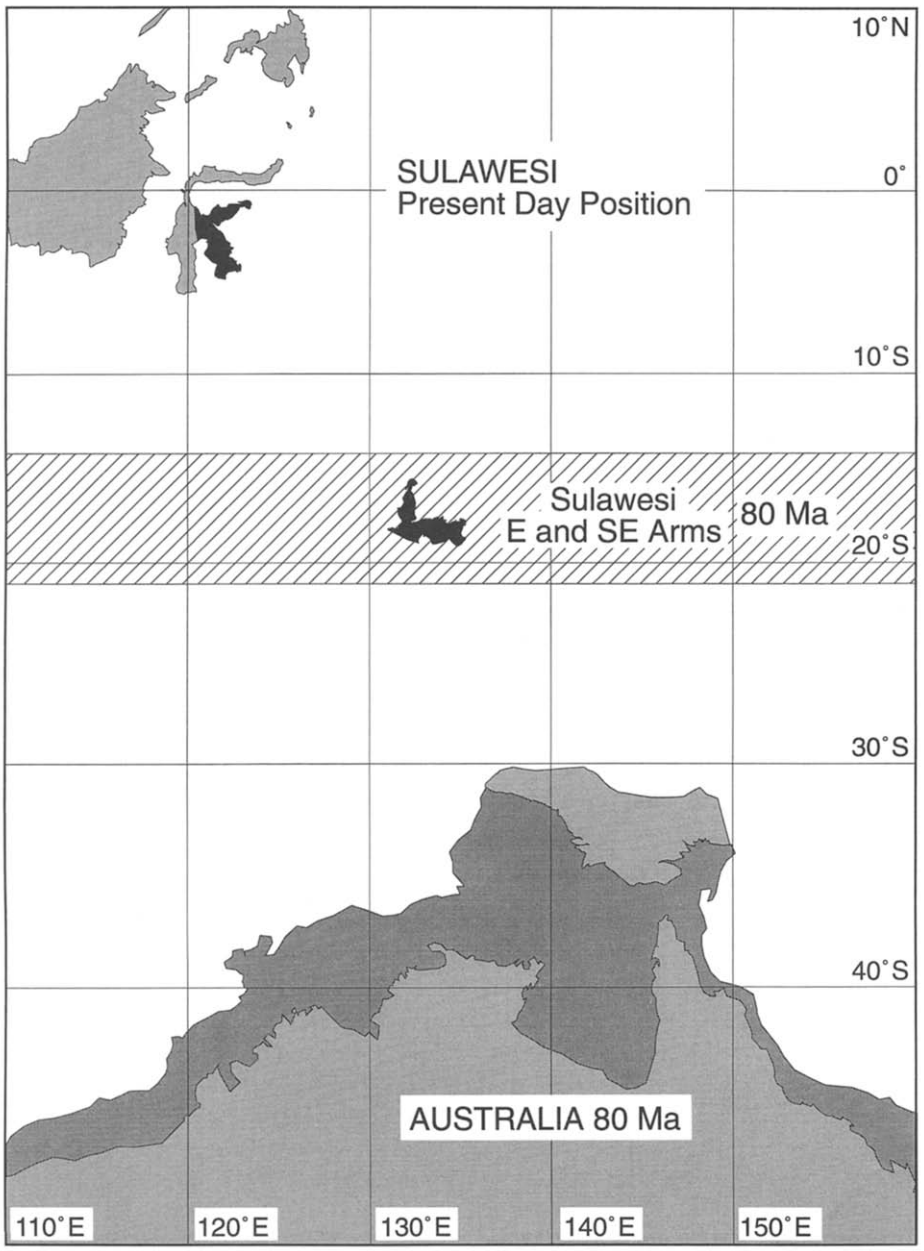


Fig. 12. Sketch map showing the probable palaeolatitude (shaded) of eastern Sulawesi relative to Australia at 80 Ma. Sulawesi is positioned on the basis of the palaeomagnetism of the Balantak ophiolite [this paper]. The palaeoposition of Australia is based on Royer and Sandwell [20]; its margin is delineated by the present 200 m bathymetric contour and the southern part of Papua New Guinea that is believed to have been contiguous. Van der Voo [29] places Australia some 10° further south than this based on Australian palaeomagnetic data alone. The present positions of the rest of Sulawesi and Borneo are shown for reference only, but palaeomagnetic data indicate that they have been located close to the equator since the Late Cretaceous.

Table 3
Palaeomagnetic directions of high-T_B components in the Boba cherts (locality 1.7°S, 121.8°E)

Site	In situ			Tilt corrected				
	N	$D°$	$I°$	Bedding strike/dip	$D°$	$I°$	k	α_{95}
SS10	6	297.5	−23.0	215/23	295.0	−45.8	13.4	19.0
SS11	5	282.0	−12.6	215/23	277.8	−33.5	41.6	12.0
SS12	5	289.4	−18.3	220/25	283.5	−41.3	21.2	17.0
SS13	9	292.0	−15.6	220/20	288.7	−34.5	21.6	11.3
B = 4	25	290.1	−17.5		285.9	−38.9 *	99.1	9.3

* Pole: 15.4°N, 9.5°E, $d_p = 6.6°$, $d_m = 11.1°$.

similar Cretaceous and Tertiary rocks in the nearby islands of the Halmahera group where palaeomagnetic results from ophiolitic and associated rocks indicate subequatorial latitudes of formation [19] and reconstructions of the Halmahera region indicate an origin within the Pacific. The palaeolatitudes of all Sulawesi rocks are significantly farther south, but north of the northern Australian margin for both the Cretaceous (Fig. 12) and the early Tertiary [20,29]. The East Sulawesi palaeolatitude is similar to those determined from Late Cretaceous–Palaeocene rocks in Misool (between Seram and the Bird's Head of New Guinea (Fig. 1)) that are consistent with those areas having been among an assemblage of microcontinents that had earlier been part of the Australian margin [29,30].

There is evidence of subduction beneath the West Arm of Sulawesi since at least the Late Cretaceous [1], and this area formed part of the east Sunda margin. The palaeolatitudes recorded by the Sulawesi East Arm rocks indicate subduction of Indian Ocean [e.g., 2] rather than Pacific lithosphere [21] beneath the Sunda margin in the late Mesozoic and early Tertiary. The East Sulawesi ophiolite and the West Arm were juxtaposed by the Early Miocene [22]. Palaeomagnetic evidence indicates little or no latitude change of Sunda and West Sulawesi since the Cretaceous [8,23,24] but significant anticlockwise rotation during the Miocene, indicating that before the Miocene the active margin faced south. Thus, a plausible scenario for the plate tectonic history of the East Arm of Sulawesi is of ophiolite formation at ~ 20°S between the Early Cretaceous and early Tertiary, followed by northward transport and large rotations of small blocks (with dimensions of just tens of kilometres). There is insufficient information to delineate the blocks. The tectonic setting of the ophiolite formation is uncertain. The East Arm ophiolite could represent Indian Ocean lithosphere and be equivalent to the surmised trapped Indian Ocean crust in the Banda Sea [e.g., 25–28] which is thought to be Cretaceous or Eocene in age. Alternatively, it could be composite and include parts of the Sunda forearc and material such as sediments, igneous rocks and seamounts scraped from the subducting Indian Ocean plate as Australia moved northwards during the Tertiary. Such a hypothesis would account for the age discrepancies and the tectonically complex character of the ophiolite.

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